STFSFormer: A Spiking Transformer with Spatial-Temporal Full-Spike Attention for EEG Emotion Recognition

***Abstract*—Electroencephalography (EEG) enables millisecond resolution affect sensing, but reliable recognition remains challenging in always-on settings: emotion cues are time-frequency transients distributed across multiple bands, while continuous deep models incur dense computation and energy overhead. We present STFSFormer, an EEG-tailored spiking framework that aligns spike dynamics and attention computation with these signal and deployment constraints. At the neuron level, we propose a High-Frequency Enhanced Leaky Integrate-and-Fire (HELIF) mechanism that injects a differential-driven synaptic path to preserve rapid EEG variations, together with a refractory trace to stabilize sparse spiking. Building on this neuron, a learnable multi-frequency spike encoder runs parallel branches with different decay profiles to extract complementary band-specific evidence and produces compact spike tokens with temporal downsampling. On top of the spike tokens, we introduce Spatial-Temporal Full-Spike Attention (STFSAtten), a block-wise spike-driven attention module that captures local temporal correlations via associative computation, preserving event-driven execution and avoiding dense attention maps. Experiments on three benchmarks (SEED, SEED-V, and DREAMER) show consistent gains over representative baselines, achieving 92.54 % accuracy on SEED, 87.63 % on SEED-V, and 85.91 %/85.47 % on DREAMER valence/arousal. STFSFormer is lightweight with 1.80M parameters and 0.46G MACs, and demonstrates favorable FPGA efficiency with 0.18 mJ per EEG segment, supporting practical deployment under resource constraints.**

***Index Terms*—Electroencephalography, Emotion recognition, Spiking Neural Networks, Transformer.**

I. Introduction

EMOTIONAL disorders, including depression and anxiety, are increasingly prevalent and constitute a major threat to global mental health [1], [2]. Automatic emotion recognition aims to infer affective states from observable signals and has attracted growing interest in psychological research and clinical practice [3], with potential for early screening, continuous monitoring, and personalized intervention. However, reliable recognition remains challenging due to the intrinsic complexity of affective processes and strong inter-individual variability [4].

Meanwhile, mental-health assessment is shifting from episodic, clinic-centered evaluation toward continuous sensing in daily life, making wearable and always-on affective monitoring a practical and timely direction. Among various sensing modalities, electroencephalography (EEG) is particularly attractive for wearable affective sensing because it directly reflects fast neural dynamics underlying affective processing [5]–[7]. Compared with invasive recordings such as electrocorticography, EEG provides a non-invasive and real-time view of cortical activity with millisecond temporal precision, which is crucial for capturing rapid and transient affective responses [8]. Moreover, EEG devices are portable, low-cost, and increasingly easy to deploy at scale [9]. A practical obstacle, however, is that wearable deployment imposes strict constraints on computation and energy. [10]

Recent progress has introduced strong deep models for EEG emotion recognition, including Transformer-style sequence modeling with various lightweight designs [11]–[15]. Yet, many efficiency-oriented solutions are primarily derived from generic compression or architectural downsizing, and are optimized from a computational perspective rather than being aligned with the signal properties and neurophysiological mechanisms of EEG [16].

From a neuro-signal perspective, affective EEG signatures are inherently time–frequency coupled. Emotion-relevant information arises from band-limited rhythms and short-lived transients, together with structured coordination across regions [5], [6], [17]. In particular, discriminative evidence can reside in rapid fluctuations and transient micro-events; overly smooth temporal processing may attenuate such cues, while dense, uniform computation can dilute them and hinder always-on deployment [10]. Thus, a deployable model should be lightweight yet neuro-signal consistent: sensitive to fast temporal changes, explicitly modeling structured temporal interactions, under resource constraints suitable for wearable and always-on settings.

This leads to a central challenge for practical EEG-based emotion recognition: how to jointly represent rapid temporal dynamics and frequency-rich transient patterns of EEG signals, while maintaining an efficiency profile compatible with always-on deployment. Addressing this challenge calls for computational primitives whose temporal processing is derived from EEG characteristics, rather than obtained solely by scaling down generic deep networks [18], [19]. To this end, we explore a spiking neural architecture that better matches both EEG mechanisms and deployment constraints.

Spiking Neural Networks (SNNs) encode information with sparse spike events and enable event-driven computation, offering a promising path toward energy-efficient inference [20]–[22]. Crucially, the above requirements are coupled: preserving transient EEG evidence typically increases temporal modeling demands, whereas always-on deployment penalizes dense continuous computation. STFSFormer is designed to address this coupled constraint directly by performing both spike generation and temporal reasoning in an event-driven manner. We propose STFSFormer, a spiking Transformer tailored for EEG-based emotion recognition, with the following contributions:

- We propose High-Frequency Enhanced Leaky Integrate-and-Fire (HELIF) spiking neuron that injects a transient-preserving time–frequency inductive bias via differential-driven synaptic dynamics, while stabilizing event-driven firing through a refractory trace. Building on this neuron,

we design a parallel multi-frequency spike encoder that extracts complementary band-specific cues.
- We design Spatial-Temporal Full-Spike Attention (STFSAtten) for EEG that captures local temporal correlations via associative computation without softmax normalization, preserving spike sparsity and enabling efficient spike-domain interaction modeling.
- We develop an end-to-end spiking architecture, STFSFormer, that integrates the proposed neuron, encoder and STFSAtten, and we validate its effectiveness and efficiency on three EEG benchmarks, including SEED [23], SEED-V [24], and DREAMER [25], demonstrating consistent accuracy gains together with favorable deployment-oriented characteristics.

II. Method

In this section, we present the details of the proposed method. The architecture of the STFSFormer model is illustrated in Fig. 1, which will be described in more detail shortly. Fig. 1 presents the overall pipeline of STFSFormer and the functionality of each component. Starting from raw EEG segments, the spike encoding module in (a) converts continuous signals into sparse spike trains while extracting informative representations; it employs a multi-frequency parallel encoding scheme to preserve band-specific dynamics and ensure that frequency-rich affective cues are sufficiently captured in the spike domain. The resulting spike tokens are then fed into our proposed STFSAtten module in (b), which performs spike-driven attention tailored to EEG and efficiently models local temporal dependencies, maintaining event-driven computation. The spike-based MLP in (c) further transforms these spike tokens through spike-domain nonlinear mapping, complementing the temporal mixing introduced by attention. After stacking such spiking blocks, the classification head in (d) aggregates the final spike representations and outputs emotion predictions. Finally, (e) explains the fusion step inside (a): the spike streams from multiple frequency branches are concatenated and projected into a unified token space, yielding a compact spiking representation that integrates complementary band information for subsequent processing.

We propose a Spiking Transformer with STFSAtten for EEG-based emotion recognition. The framework comprises five main components: (a) a patch embedding module that transforms multi-channel EEG signals into spike-based representations via convolutional layers and multi-step LIF neurons; (b) L transformer blocks, each integrating our STFSAtten mechanism and a spike-based MLP; and (c) a classification head for emotion prediction.

Unlike spatial-only attention in existing spiking transformers, STFSAtten captures both intra-temporal and cross-channel correlations within each block, enabling comprehensive spatiotemporal feature learning while maintaining the binary, event-driven computation paradigm of SNNs.

A. EEG Representation

To obtain a unified representation across datasets and suppress both slow drifts and high-frequency noise, we first apply an offline zero-phase band-pass filter of 0.5–50 Hz to all EEG signals. This range covers the major rhythm bands (δ, θ, α, β, and lower γ) that are widely reported to be associated with affective processing, while attenuating baseline wander, muscle artifacts, and sensor-level noise. [26] After filtering, we perform channel-wise z-score normalization on each trial to mitigate inter-subject and inter-session variability, so that subsequent modeling focuses on within-trial spatiotemporal patterns rather than global amplitude differences.

For temporal modeling, we first segment each continuous EEG recording into fixed-length windows and treat each window as an independent sample. Accordingly, each sample is represented as $\mathbf{X} \in \mathbb{R}^{C\times T}$, where C is the number of EEG channels and T is the number of time points per window; both are dataset-dependent and follow the corresponding acquisition protocol.

During training, we additionally apply data augmentation to improve robustness and generalization: with probability 0.3, one of the following perturbations is randomly selected and applied to a segment—additive Gaussian noise ($\sigma = 0.05$), random amplitude scaling with a factor sampled from $[0.95, 1.05]$, temporal jittering within ± 20 ms, or Mixup regularization [27] with $\alpha = 0.2$.

B. High-Frequency Enhanced Spiking Neuron

EEG captures cortical population activity with millisecond resolution, where affective cues are often expressed as a mixture of band-limited rhythms and short-lived transients spanning multiple frequency ranges. Such cues are inherently time–frequency coupled: reliable recognition benefits from short-term temporal context, yet it also requires preserving rapid fluctuations that may carry discriminative emotion-related information [28]. Therefore, spiking neurons for EEG-based affective modeling should simultaneously provide (i) temporal dependency modeling and (ii) sensitivity to fast variations, while maintaining stable and sparse event-driven activations for efficient deployment.

In conventional SNNs, the Leaky Integrate-and-Fire (LIF) neuron is the fundamental computational unit [21], with discrete-time dynamics

$$\begin{aligned} v[t] &= \eta v[t-1] + x[t] - V_{\mathrm{th}} s[t-1] \\ s[t] &= H_{\mathrm{sg}}(v[t] - V_{\mathrm{th}}) \end{aligned} \tag{1}$$

where $x[t]$ is the input current, $v[t]$ the membrane potential, $s[t]$ the binary spike output, V_{th} the firing threshold, and $\eta = e^{-\Delta t/\tau_m}$ the leakage factor determined by the membrane time constant τ_m and time step Δt. $H_{\mathrm{sg}}(\cdot)$ denotes a surrogate Heaviside function that preserves step-like spiking in the forward pass while enabling gradient-based learning [29].

From a signal-processing viewpoint, the recursion in (1) behaves like an exponential smoother, it integrates evidence over time and is effective for slowly varying components, but it also suppresses abrupt micro-events and rapid fluctuations, making the neuron less responsive to transient, time-localized cues. A straightforward remedy is to reduce the membrane time constant or increase spiking depth to boost responsiveness; however, these changes often destabilize firing, amplify

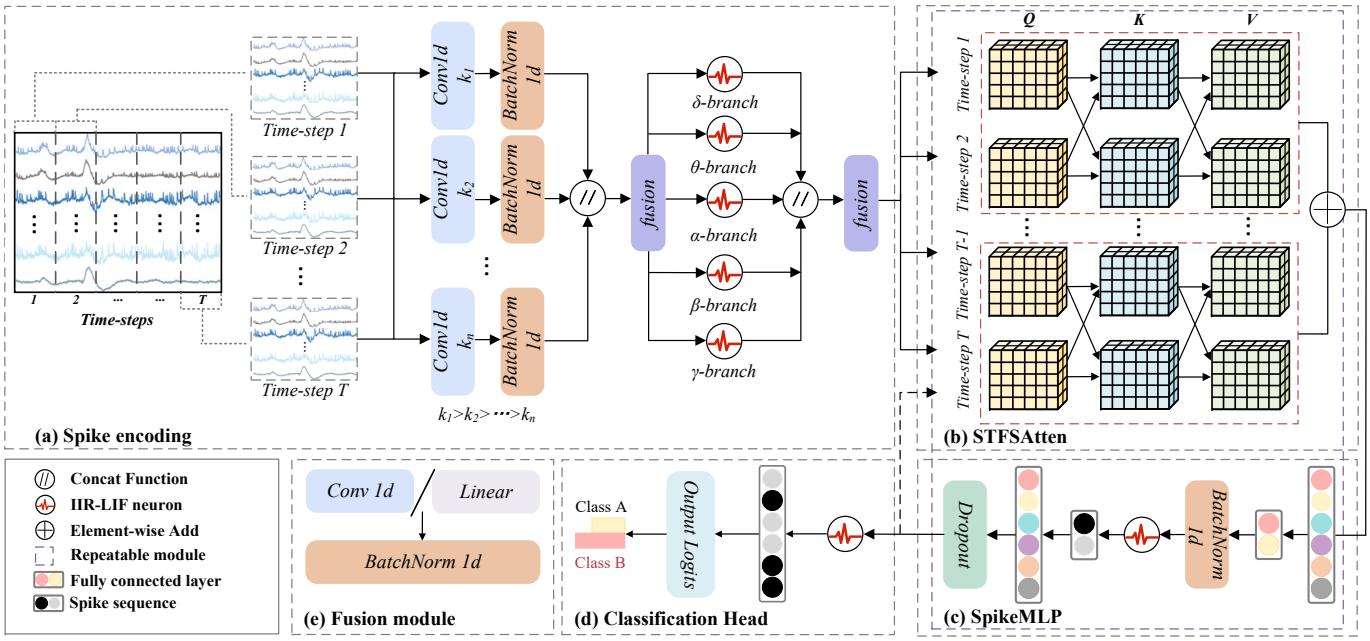


Fig. 1: The overall architecture of the proposed STFSFormer for emotion recognition.

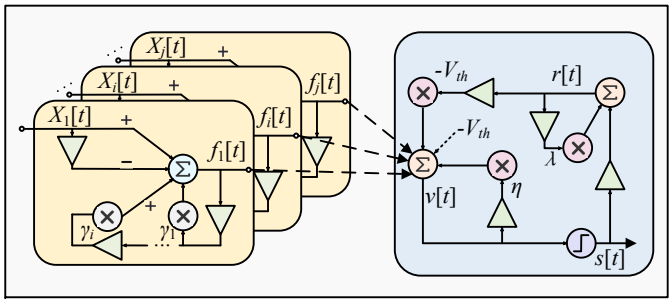


Fig. 2: Overview of the proposed HELIF neuron. By introducing a differential-driven synaptic path and a refractory trace, the neuron enhances sensitivity to rapid EEG variations while stabilizing spiking activity for efficient event-driven processing.

noise-driven spikes, or weaken the temporal context that EEG decoding relies on. This motivates a more structured modification that enhances transient sensitivity at the synaptic dynamics level while retaining controllable short-term memory and stable sparse spiking.

We propose a HELIF neuron, as Fig. 2 shows, that explicitly amplifies rapid variations while retaining lightweight recurrent memory. Specifically, we introduce a differential-driven synaptic path:

$$f[t] = \gamma f[t-1] + (x[t] - x[t-1]), \tag{2}$$

where $\gamma \in (0,1)$ controls the decay of the feedback term $f[t-1]$. The differential component $(x[t] - x[t-1])$ serves as a discrete-time high-pass drive: it suppresses slow trends while emphasizing abrupt temporal changes, thereby improving sensitivity to transient EEG variations that may carry emotion-relevant cues. Meanwhile, the recurrent term $\gamma f[t-1]$ preserves short-term context, preventing the response from becoming purely instantaneous.

A direct differential drive may increase spiking volatility. To stabilize event-driven firing and encourage sparse, regularized spike patterns, we further introduce a refractory trace:

$$\begin{aligned} v[t] &= \eta v[t-1] + f[t] - V_{\text{th}} r[t-1], \\ r[t] &= \lambda r[t-1] + s[t], \\ s[t] &= H_{\text{sg}}(v[t] - V_{\text{th}}), \end{aligned} \tag{3}$$

where $r[t]$ accumulates recent spikes with decay coefficient $\lambda \in (0,1)$. The term $V_{\text{th}} r[t-1]$ acts as short-term self-inhibition, which counterbalances the transient-enhanced excitation induced by (2). This mechanism improves stability during training and yields sparse spiking activity that is favorable for event-driven computation.

The proposed HELIF neuron remains lightweight and hardware-friendly: it only introduces two auxiliary states ($f[t]$ and $r[t]$) and uses simple add–subtract–multiply recursions without expensive operations. In our implementation, the neuron states are reset at the beginning of each EEG window, so the dynamics are sequence-local and do not leak across segments, which matches the windowed EEG setting used throughout this paper. By tuning γ and λ, HELIF flexibly balances transient emphasis and short-term memory, providing an EEG-aligned time–frequency inductive bias while preserving end-to-end trainability with surrogate gradients.

C. Spike Encoding

Thus, an appropriate encoding mechanism is required to transform preprocessed EEG segments into spike sequences. Classical neural coding schemes such as rate coding and delay coding represent information by firing frequency or inter-spike intervals, but their coding rules are usually fixed and not optimized jointly with the downstream task. This limitation is particularly pronounced for EEG, where emotion-relevant information is often carried by rapid oscillatory components and transient events. Inspired by learnable spike encoders [30], we adopt a fully trainable encoding scheme that maps each

multichannel EEG segment to a set of input currents and then converts them into spikes through the proposed high-frequency–enhanced spiking neuron.

At each time step t, the corresponding input vector $\mathbf{X}^{(t)} \in \mathbb{R}^C$ is first projected into a latent current space by a learnable linear transformation. The resulting current sequence $\{\mathbf{x}^{(t)}\}_{t=1}^{T}$ is then fed into a bank of high-frequency–enhanced HELIF neurons. $\gamma, \eta, \lambda \in (0,1)$ are decay coefficients controlling temporal feedback; V_{th} is the firing threshold; and $H_{\text{sg}}(\cdot)$ is a surrogate Heaviside function. The differential term $\mathbf{x}^{(t)} - \mathbf{x}^{(t-1)}$ acts as a discrete high-pass operator, driving the encoder to respond preferentially to rapid temporal changes in the input. In this way, the encoder jointly learns the projection $\mathbf{W}_{\text{enc}}$ and the neuron dynamics to produce spike trains $\{\mathbf{s}^{(t)}\}$ that emphasize high-frequency EEG components, providing a task-adaptive and hardware-friendly spike representation for subsequent spiking layers.

Raw EEG is a continuous-valued time series, whereas SNNs operate on discrete spike events. In affective EEG analysis, discriminative temporal patterns are often non-stationary and vary across emotional states and subjects, making fixed spike generation rules difficult to apply consistently [31]. We therefore adopt a learnable spike encoding scheme that converts each windowed EEG segment into spike-based tokens in an end-to-end manner, enabling task-adaptive representation learning for subsequent spiking modules.

Formally, let $\mathbf{X} \in \mathbb{R}^{C\times T}$ denote an EEG segment with C channels and T time steps. Instead of directly spiking the raw signals, we first extract compact temporal representations using a multi-scale temporal encoder:

$$\mathbf{H} = \Phi_{\text{ms}}(\mathbf{X}), \quad \mathbf{H} \in \mathbb{R}^{T'\times D}, \tag{4}$$

where $\Phi_{\text{ms}}(\cdot)$ produces a feature sequence of length T' and dimension D. Besides capturing multi-scale temporal patterns, $\Phi_{\text{ms}}(\cdot)$ can downsample the temporal axis, reducing the number of time steps processed by subsequent spiking and attention blocks.

a) Multi-frequency spiking encoder: Emotion-related EEG cues are distributed across multiple oscillatory bands and temporal scales. To capture complementary dynamics without relying on fixed, handcrafted frequency decomposition, we employ K parallel spiking encoding branches with distinct temporal response profiles. Each branch applies the HELIF neuron to the same feature stream $\mathbf{H}$, yielding

$$\mathbf{S}^{(k)} = \Psi_k(\mathbf{H}), \quad \mathbf{S}^{(k)} \in \{0,1\}^{T'\times D}, \quad k = 1,\dots,K, \tag{5}$$

where $\Psi_k(\cdot)$ denotes the neuron dynamics with branch-specific parameters, and $\mathbf{S}^{(k)}$ is the corresponding spike sequence.

b) Spike fusion: We concatenate the K spike streams along the feature dimension and project them back to the model space:

$$\begin{aligned} \tilde{\mathbf{S}} &= \text{Concat}\big(\mathbf{S}^{(1)},\dots,\mathbf{S}^{(K)}\big) \in \{0,1\}^{T'\times KD}, \\ \mathbf{Z} &= \text{BN}\Big(\tilde{\mathbf{S}}\mathbf{W}_{\text{fuse}}\Big) \in \mathbb{R}^{T'\times D}, \end{aligned} \tag{6}$$

where $\mathbf{W}_{\text{fuse}} \in \mathbb{R}^{KD\times D}$ is a learnable fusion matrix and $\text{BN}(\cdot)$ denotes batch normalization. The resulting $\mathbf{Z} \in \mathbb{R}^{T'\times D}$ serves as the spike-based token sequence for subsequent spike-driven attention and spiking MLP blocks. During training, surrogate gradients are used for spike generation to enable backpropagation while preserving binary spikes in the forward pass.

D. Spatial-Temporal Full-Spike Attention

In practice, all operations are applied in parallel across a mini-batch. Capturing local temporal interactions is crucial for affective EEG decoding, and attention provides a principled way to model such dependencies [32]. However, standard full attention is impractical for spike-based EEG sequences due to its quadratic cost and dense softmax normalization, which conflicts with sparse, event-driven spike representations [33], [34]. Given the spike-based token sequence produced by the spike encoder, we adopt STFSAtten, a block-wise spike-driven attention mechanism, that enforces temporal locality and avoids materializing dense attention maps.

a) Spike-Conditioned Projections: Let $\mathbf{X} \in \mathbb{R}^{T'\times D}$ denote the spike-based token sequence output by the spike encoder, where T' is the downsampled number of time steps and D is the embedding dimension. Query, key, and value representations are generated via linear projections followed by the HELIF spiking operator $\phi(\cdot)$ (trained with surrogate gradients):

$$\begin{aligned} \mathbf{Q} &= \phi(\mathbf{X}\mathbf{W}_Q), \\ \mathbf{K} &= \phi(\mathbf{X}\mathbf{W}_K), \\ \mathbf{V} &= \phi(\mathbf{X}\mathbf{W}_V), \end{aligned} \tag{7}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D\times D}$. We reshape them into a multi-head format:

$$\begin{aligned} \mathbf{Q},\mathbf{K},\mathbf{V} &\in \mathbb{R}^{N_h\times T'\times d_h}, \\ d_h &= \frac{D}{N_h}, \end{aligned} \tag{8}$$

where N_h is the number of heads and d_h is the per-head dimension.

b) Local Temporal Blocking: To balance modeling capacity and efficiency, we partition the temporal axis into $N_c = T'/L$ non-overlapping blocks of size L. For the i-th block, we extract

$$\begin{aligned} \mathbf{Q}^{(i)} &= \mathbf{Q}[:, iL:(i+1)L, :], \\ \mathbf{K}^{(i)} &= \mathbf{K}[:, iL:(i+1)L, :], \\ \mathbf{V}^{(i)} &= \mathbf{V}[:, iL:(i+1)L, :], \end{aligned} \tag{9}$$

where $\mathbf{Q}^{(i)}, \mathbf{K}^{(i)}, \mathbf{V}^{(i)} \in \mathbb{R}^{N_h\times L\times d_h}$. This design focuses attention on local temporal neighborhoods, which are especially relevant for EEG oscillations and transient affective responses, while preventing the quadratic growth in memory associated with long sequences.

c) Associative Spike Attention: Within each temporal block, attention is computed in a Softmax-free manner by exploiting associativity to avoid forming a dense $L \times L$ attention map:

$$\begin{aligned} \mathbf{A}^{(i)} &= (\mathbf{K}^{(i)})^{\top}\mathbf{V}^{(i)} \in \mathbb{R}^{N_h\times d_h\times d_h}, \\ \mathbf{O}^{(i)} &= \mathbf{Q}^{(i)}\mathbf{A}^{(i)} \cdot \alpha \in \mathbb{R}^{N_h\times L\times d_h}, \end{aligned} \tag{10}$$

where $\alpha = 1/\sqrt{d_h}$ is the scaling factor. Compared with explicitly computing $\mathbf{Q}^{(i)}(\mathbf{K}^{(i)})^\top$, the intermediate interaction is compressed to $d_h \times d_h$ per head, yielding a more memory-efficient computation while retaining expressive temporal mixing within each block.

The outputs from all blocks are concatenated along the temporal dimension:

$$\mathbf{O}_{\text{cat}} = \text{Concat}\big(\{\mathbf{O}^{(i)}\}_{i=0}^{N_c-1}\big) \in \mathbb{R}^{N_h \times T' \times d_h}. \tag{11}$$

d) Event-Driven Readout and Residual Mapping: To maintain spike-based activations throughout the network, we apply an HELIF layer to the concatenated output and then merge heads:

$$\begin{aligned} \mathbf{O}_s &= \phi(\mathbf{O}_{\text{cat}}) \in \mathbb{R}^{N_h \times T' \times d_h}, \\ \mathbf{U} &= \text{Merge}(\mathbf{O}_s) \in \mathbb{R}^{T' \times D}. \end{aligned} \tag{12}$$

Finally, we apply a learnable output projection with batch normalization and a residual connection:

$$\begin{aligned} \text{STFSAtten}(\mathbf{X}) &= \text{BN}(\mathbf{U}\mathbf{W}_O), \\ \mathbf{Y} &= \mathbf{X} + \text{STFSAtten}(\mathbf{X}), \end{aligned} \tag{13}$$

where $\mathbf{W}_O \in \mathbb{R}^{D \times D}$ and $\text{BN}(\cdot)$ stabilizes spike statistics during training.

e) Complexity: Within each block, the proposed computation avoids materializing a dense $L \times L$ attention matrix. Overall complexity scales linearly with the sequence length T', while depending quadratically only on the per-head feature dimension through the compact $d_h \times d_h$ interaction, providing an efficient trade-off for long EEG sequences.

III. Results

A. Experimental Setup

All experiments are implemented in PyTorch and conducted on a single NVIDIA RTX 4090 GPU. We evaluate the proposed STFSFormer on three representative EEG-based emotion recognition benchmarks: SEED, SEED-V, and DREAMER. SEED and SEED-V contain multi-channel EEG recordings collected under visual emotional stimuli with discrete emotion labels, while DREAMER provides multi-channel EEG together with continuous self-reported valence and arousal scores, from which we construct two binary classification tasks (DREAMER-Valence and DREAMER-Arousal) by thresholding the ratings. For all datasets, we adopt a subject-dependent splitting strategy, where the data of each subject are divided into training, validation, and test sets with a ratio of 8:1:1. Unless otherwise specified, all reported performance metrics are computed under this partitioning protocol.

B. Model Performance Analysis

We conduct a sensitivity study on four hyperparameters that directly affect spike-based temporal modeling: the attention chunk size L, the multi-branch decay span Δ_γ in the spike encoder, the shared decay γ_{shared} used by later spiking neurons, and the temporal downsampling factor s. As shown in Fig. 3, all four datasets exhibit consistent optima at L=3, Δ_γ=0.16, γ_{shared}=0.89, and s=2. Overall, the curves reveal a common mechanism: accurate affect decoding requires a matched temporal receptive field for transient, band-specific EEG cues. With too small L, the spike-driven attention cannot reliably integrate short-range oscillatory structure, yielding fragmented evidence over time. With overly large L or an aggressive Δ_γ, the effective temporal integration becomes excessively inertial, attenuating abrupt state transitions and smearing discriminative transients that often characterize affective responses.

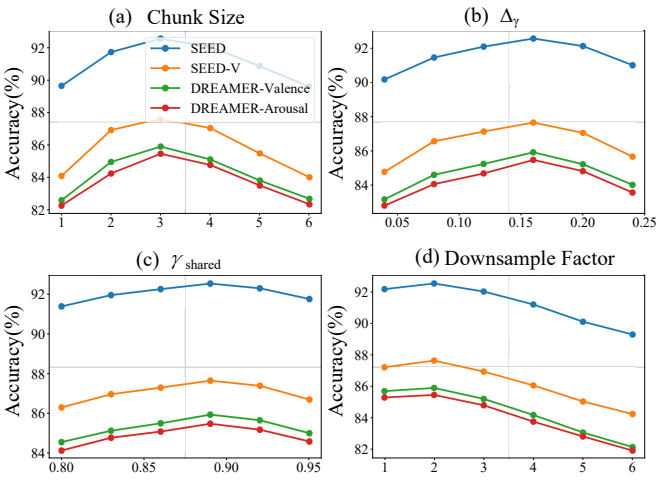


Fig. 3: Hyperparameter sensitivity of STFSFormer on four EEG emotion recognition benchmarks. We sweep (a) the attention chunk size L, (b) the multi-branch decay span Δ_γ in the spike encoder, (c) the shared decay γ_{shared} for later spiking neurons, and (d) the temporal downsampling factor s. Across datasets, the best performance consistently occurs at L=3, Δ_γ=0.16, γ_{shared}=0.89, and s=2.

Likewise, γ_{shared} peaks around 0.89, suggesting that downstream spiking units need a balanced memory–responsiveness regime: sufficient persistence to accumulate weak but consistent patterns, yet enough reactivity to preserve fast deviations. For temporal downsampling, s>2 monotonically degrades performance, consistent with the notion that excessive compression discards short-lived events and disrupts the temporal alignment required by spike-based dynamics. Based on these observations, we adopt L=3, Δ_γ=0.16, γ_{shared}=0.89, and s=2 as the default configuration for all experiments.

C. Additional Sensitivity Studies

To further substantiate the mechanism design beyond the main hyperparameter sweep in Fig. 3, we conduct two additional sensitivity studies that target spiking dynamics and multi-band evidence aggregation. We analyze the refractory decay λ in HELIF, which governs spike-history-dependent self-inhibition, and the number of parallel branches K in the multi-frequency spike encoder, which controls how densely the model samples complementary time–frequency scales. Tables II and III report representative results across all benchmarks.

Across datasets, performance peaks at an intermediate-to-high refractory decay, with λ=0.90 consistently giving the best accuracy. This pattern matches the role of the refractory trace in HELIF. When λ is too small, self-inhibition decays rapidly and the differential-driven excitation is less regulated,

TABLE I: Performance comparison on three EEG emotion recognition datasets. (Mean (%) ± Standard Deviation (%))

Method	SEED		SEED-V		DREAMER (Valence)		DREAMER (Arousal)	
	ACC	F1	ACC	F1	ACC	F1	ACC	F1
DCNN [35]	75.39 ± 3.21	74.97 ± 3.45	66.19 ± 4.82	63.94 ± 5.17	63.49 ± 5.34	66.03 ± 5.08	70.24 ± 4.67	65.80 ± 5.22
EEGNet [36]	84.36 ± 2.45	84.04 ± 2.68	72.44 ± 3.87	72.20 ± 4.12	62.55 ± 4.95	63.23 ± 4.78	61.17 ± 5.03	62.69 ± 4.91
GAT [37]	92.48 ± 1.78	81.86 ± 2.95	81.45 ± 3.12	81.23 ± 3.34	79.89 ± 3.45	79.45 ± 3.67	80.12 ± 3.38	79.78 ± 3.52
MMP-GCN [38]	82.13 ± 2.67	81.86 ± 2.89	79.32 ± 3.25	78.94 ± 3.48	77.83 ± 3.56	77.46 ± 3.79	77.24 ± 3.61	76.87 ± 3.84
Brain-GRL [39]	88.08 ± 2.12	87.84 ± 2.34	84.32 ± 2.87	80.41 ± 3.42	84.31 ± 2.95	83.10 ± 3.18	85.27 ± 2.78	87.05 ± 2.41
Spikformer [40]	83.24 ± 2.54	82.94 ± 2.76	80.43 ± 3.08	80.09 ± 3.31	78.90 ± 3.42	78.57 ± 3.65	78.31 ± 3.49	77.95 ± 3.72
SpikingGCN [41]	92.01 ± 1.82	91.99 ± 1.85	84.46 ± 2.79	84.54 ± 2.81	82.12 ± 3.12	82.87 ± 2.96	83.38 ± 3.05	83.82 ± 2.88
SCNN [42]	88.44 ± 2.18	90.09 ± 1.96	81.54 ± 3.14	81.00 ± 3.37	79.32 ± 3.48	79.05 ± 3.71	80.68 ± 3.41	80.61 ± 3.53
TCJA-SNN [43]	86.89 ± 2.31	85.78 ± 2.56	78.12 ± 3.42	77.67 ± 3.65	76.34 ± 3.67	76.12 ± 3.89	76.78 ± 3.73	76.45 ± 3.95
FSTA-SNN [44]	89.12 ± 2.05	88.67 ± 2.28	79.56 ± 3.29	79.23 ± 3.52	77.12 ± 3.62	77.68 ± 3.48	77.98 ± 3.56	77.68 ± 3.63
STFSFormer (Ours)	**92.54 ± 1.68**	**90.18 ± 1.92**	**87.63 ± 2.34**	**87.25 ± 2.47**	**85.91 ± 2.56**	**85.57 ± 2.69**	**85.47 ± 2.61**	**85.11 ± 2.74**

TABLE II: Sensitivity to the refractory decay λ in HELIF. Accuracy is reported in %.

λ	SEED	SEED-V	DREAMER-V	DREAMER-A
0.70	91.98	86.91	85.21	84.84
0.75	92.21	87.19	85.47	85.05
0.80	92.39	87.43	85.68	85.26
0.85	92.50	87.56	85.82	85.39
0.90	**92.54**	**87.63**	**85.91**	**85.47**
0.95	92.31	87.28	85.62	85.14

TABLE III: Sensitivity to the number of encoder branches K. Accuracy is reported in %.

K	SEED	SEED-V	DREAMER-V	DREAMER-A
1	91.62	86.31	85.03	84.71
2	92.05	86.84	85.41	85.09
3	92.33	87.21	85.63	85.27
4	92.47	87.46	85.80	85.36
5	**92.54**	**87.63**	**85.91**	**85.47**
6	92.49	87.52	85.86	85.41

which can lead to noisier spike bursts and less stable evidence accumulation. When λ is too large, inhibition persists excessively and suppresses transient-sensitive responses, weakening the model's ability to preserve short-lived discriminative fluctuations. The optimum around λ=0.90 indicates a balanced regime where spikes remain sparse and stable while the neuron still responds promptly to affect-related micro-events.

Varying K yields a clear saturation trend. Increasing K from 1 to 4 steadily improves accuracy, suggesting that additional parallel branches provide complementary evidence across distinct temporal scales and help preserve transient cues that may be concentrated in specific oscillatory regimes. Notably, the best performance is consistently achieved at K=5, which is also physiologically compatible with the canonical partition of EEG activity into five major rhythm bands (δ, θ, α, β, and low-γ). While our encoder does not enforce a hard band-pass decomposition, assigning five branches with different decay profiles effectively allocates capacity to multiple band-dominant dynamics and their short-lived transients, yielding a more complete coverage of affect-related spectral structure. The gain becomes marginal beyond K=5, and a slight drop at K=6 is consistent with redundancy in scale sampling and more correlated spike streams after fusion, which can dilute discriminative band-specific evidence. Based on these observations, we use K=5 as the default configuration, matching both the empirical optimum and the neurophysiological intuition of multi-band EEG organization.

D. Comparison with Existing Methods

Table I compares STFSFormer with representative dense EEG models and recent spike-based architectures under the same evaluation protocol. Overall, STFSFormer achieves the highest accuracy on SEED, SEED-V, DREAMER-Valence, and DREAMER-Arousal, while maintaining strong F1 scores and relatively small standard deviations. These results indicate that tailoring both neuron dynamics and attention computation to the spike domain yields competitive discriminative power without relying on dense normalization or quadratic attention maps, which is desirable for resource-constrained EEG deployment.

On SEED, STFSFormer reaches 92.54 % accuracy and 90.18 % F1, matching or exceeding the best dense baseline reported in Table I and also improving over the strongest spiking baseline. This suggests that the proposed spike-driven temporal modeling can retain fine-grained discriminability even on datasets where strong dense graph models already perform competitively. On the more challenging SEED-V benchmark, STFSFormer achieves 87.63 % accuracy and 87.25 % F1, establishing a clear margin over both the best dense baseline and the strongest spiking baseline in Table I. The larger improvement on SEED-V is consistent with the motivation that fine-grained affect categories depend more heavily on transient and short-range temporal evidence, for which the proposed block-wise spike-driven attention provides a better inductive bias and a more efficient interaction form than standard dense attention.

On DREAMER, STFSFormer continues to show consistent advantages across both affective dimensions. For valence, it attains 85.91 % accuracy and 85.57 % F1, exceeding the best dense baseline and offering a larger gain over the strongest spiking baseline. For arousal, it achieves the highest accuracy of 85.47 % together with a competitive F1 of 85.11 %, remaining superior to the spiking baselines in Table I. Taken together, the comparisons across datasets and model families demonstrate that STFSFormer provides robust improvements over dense counterparts and modern spiking approaches, validating the effectiveness of combining transient-sensitive spike

dynamics with efficient spike-domain temporal attention for EEG emotion recognition.

E. Ablation Study

To examine the roles of the proposed spiking neuron design and the attention formulation, we perform controlled ablations in Table IV. Exp. 1 is the complete STFSFormer, combining the HELIF neuron with the proposed full-attention module. Exp. 2 keeps the same attention module but replaces HELIF with a standard LIF neuron. Exp. 3 retains HELIF while switching the attention to a spatial-only spiking self-attention (SSA). Exp. 4 removes both components, resulting in a baseline spiking transformer variant with LIF neurons and SSA.

Across all benchmarks, Exp. 1 delivers the strongest results. Relative to the minimal configuration in Exp. 4, the complete model improves accuracy by 9.30, 7.20, 7.01, and 7.16 percentage points on SEED, SEED-V, DREAMER-Valence, and DREAMER-Arousal, respectively. The corresponding F1 improvements are also substantial, reaching 7.24, 7.16, 7.00, and 7.16 points. These consistent gains indicate that the proposed neuron and attention components jointly provide a meaningful increase in discriminative capacity for EEG emotion recognition.

The contribution of the spiking neuron design can be isolated by comparing Exp. 1 and Exp. 2 under the same attention mechanism. Replacing HELIF with a standard LIF leads to a uniform degradation of roughly 2–3 points in both accuracy and F1 across all tasks, showing that the HELIF dynamics provide a tangible advantage for representing EEG sequences. The effect of the attention formulation is reflected by Exp. 1 versus Exp. 3: replacing the proposed full-attention with SSA results in larger drops, typically around 5–7 points on both accuracy and F1 across datasets, suggesting that the attention design is the primary driver of the performance gains. In addition, Exp. 2 consistently outperforms Exp. 3, indicating that, in this model family, the attention formulation contributes more than the neuron replacement when considered in isolation. Overall, the ablations support that the two components are complementary: the attention module strengthens temporal dependency modeling at the architectural level, while HELIF further enhances spike-based temporal representation, and their combination yields the best and most stable performance.

F. Confusion Matrix Analysis

We further inspect confusion matrices to reveal how errors distribute across affective states. Across SEED, SEED-V, and DREAMER, misclassifications are highly structured: errors concentrate on semantically and physiologically adjacent classes, while distant class pairs remain comparatively clean. This suggests that the remaining errors are largely associated with intrinsic overlap in EEG signatures across neighboring emotional conditions, rather than random confusion. From a neurophysiological perspective, such confusion patterns are consistent with prior observations that neighboring emotions

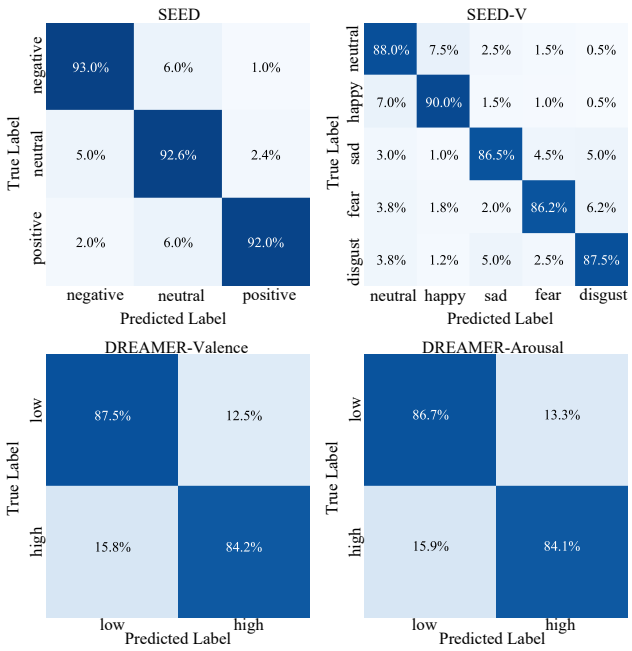


Fig. 4: Row-normalized confusion matrices (%) of STFSFormer on SEED, SEED-V, and DREAMER (valence/arousal), highlighting class-wise recognition reliability and dominant confusions between affectively adjacent states.

often share partially overlapping oscillatory regimes and cross-region coordination, and the most discriminative cues can be transient, localized in time, and spatially distributed.

Consequently, borderline samples especially in DREAMER where continuous ratings are thresholded tend to drive asymmetric false positives/negatives near the decision boundary. These observations align with the design of STFSFormer: the high-frequency–enhanced spiking dynamics prioritize rapid variations that carry affective micro-events, while the block-wise spike-driven attention preserves locally coherent temporal interactions under event-driven sparsity. Overall, the confusion matrices indicate that remaining errors largely stem from non-separable neuro-signal neighborhoods, and the proposed mechanism reduces confusion where discriminative information relies on fast, distributed spatiotemporal patterns.

G. Convergence Analysis

We analyze the optimization behavior of STFSFormer to verify that the proposed spike-based design can be trained reliably. As shown in Fig. 5, across all benchmarks, training shows consistent convergence: the objective decreases steadily after a short warm-up, validation accuracy rises and saturates, and no run exhibits divergence or persistent oscillation. The reported results are selected by the best validation epoch under the same protocol, and the small run-to-run variance is consistent with stable, repeatable convergence rather than occasional favorable runs.

This behavior aligns with the model's mechanism design. Surrogate gradients provide smooth update signals through spike generation, while HELIF combines a transient-sensitive differential synaptic pathway with a refractory trace that regularizes firing and reduces noisy, bursty updates in early

TABLE IV: Performance Analysis of Ablation Experiments [%].

Case	Spiking Neuron	Attention	Accuracy				F1-Score			
			SEED	SEED-V	Valence	Arousal	SEED	SEED-V	Valence	Arousal
(Exp.1)	HELIF	STFSAtten	**92.54**	**87.63**	**85.91**	**85.47**	**90.18**	**87.25**	**85.57**	**85.11**
(Exp.2)	LIF	STFSAtten	90.32	84.78	82.95	82.53	87.89	84.12	82.49	82.06
(Exp.3)	HELIF	SSA	87.45	81.23	79.42	79.00	84.56	80.34	79.01	78.58
(Exp.4)	LIF	SSA	83.24	80.43	78.90	78.31	82.94	80.09	78.57	77.95

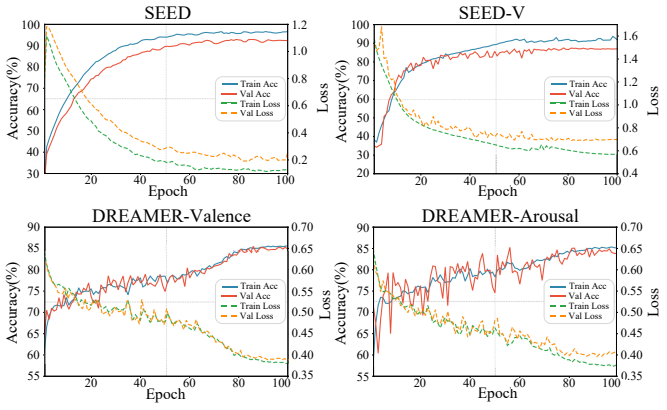


Fig. 5: Training and validation accuracy and loss curves of STFSFormer on SEED, SEED-V, and DREAMER (valence/arousal), showing stable convergence and well-behaved optimization across all benchmarks.

TABLE V: Model complexity comparison under a standard input setting.

Method	Params (M)	MACs (G)	FPS (4090)
EEGNet	0.10	0.02	2100
SpikingGCN	2.42	0.85	520
SCNN	3.17	1.10	480
STFSFormer	1.80	0.46	900

TABLE VI: FPGA resource utilization and estimated energy per EEG segment under a consistent synthesis-based evaluation protocol.

Method	LUTs (%)	BRAM (%)	DSP (%)	Power (W)	Energy (mJ)
EEGNet	58.9	22.4	48.0	2.3	4.6
SpikingGCN	62.7	31.6	18.5	1.8	3.0
SCNN	55.3	27.8	12.0	1.9	2.7
STFSFormer	45.3	24.0	8.0	1.2	1.5

training. Residual mappings and normalization further stabilize spike statistics when stacking blocks. In addition, the block-wise spike-driven attention avoids dense softmax normalization and large attention maps, yielding more predictable numerical behavior and lowering intermediate memory pressure. Together, these factors support robust end-to-end optimization of STFSFormer across datasets.

IV. Discussion

A. Model Complexity and Efficiency

Beyond recognition accuracy, STFSFormer is designed to balance model capacity and computational efficiency for practical EEG deployment. Table V reports the parameter size, multiply–accumulate operations (MACs), and throughput under a unified input setting: batch size = 1, C=62 EEG channels, T=200 time steps (1 second at 200 Hz). STFSFormer uses 1.80M parameters and 0.46G MACs while achieving 900 FPS on a single RTX 4090. Compared to EEGNet's ultra-minimal architecture, STFSFormer employs a more expressive design to capture complex spatiotemporal EEG dynamics, yielding a substantial accuracy improvement: 92.54% vs. 84.36% on SEED—an 8.18 percentage point gain. Notably, STFSFormer remains significantly more efficient than other competitive methods: it uses 25% fewer parameters than SpikingGCN and 42% fewer than SCNN, while achieving higher accuracy and 1.7× better throughput than these spiking baselines.

a) Linear-Complexity Attention: This favorable efficiency–accuracy trade-off stems from two architectural innovations. First, standard self-attention incurs $\mathcal{O}(N_h T'^2 d_h)$ complexity due to explicit computation of $\mathbf{QK}^\top$ over all T' tokens. STFSFormer avoids this quadratic cost through block-wise associative computation: within each block of size L, we compute the compact interaction $\mathbf{K}^\top\mathbf{V} \in \mathbb{R}^{d_h \times d_h}$ at cost $\mathcal{O}(N_h L d_h^2)$, then multiply by $\mathbf{Q}$ at the same cost. Across T'/L blocks, total complexity becomes $\mathcal{O}(N_h T' d_h^2)$—linear in sequence length. With d_h=64 and T'=100 (after downsampling), we store only 64×64 matrices rather than 100×100 attention maps, eliminating the $\mathcal{O}(T'^2)$ memory overhead.

b) Event-Driven Sparsity: Second, binary spike representations enable event-driven computation. Matrix products reduce to spike-gated accumulations, bypassing floating-point multiplications and removing the costly softmax normalization. Measured on SEED-V, the average spike rate is 18% across encoder and attention layers, meaning over 80% of operations are skipped in event-driven execution. Combined with the linear-complexity design, this enables STFSFormer to achieve strong accuracy while maintaining practical efficiency for resource-constrained EEG applications.

B. Energy and Hardware Considerations

To evaluate deployment on resource-constrained devices, we implement an field-programmable gate array (FPGA) prototype and summarize post-implementation resource usage and energy cost in Table VI. All designs are evaluated under the same clock constraint and streaming inference setting; power is estimated in steady state, and energy is normalized per EEG segment using the achieved segment throughput.

The observed efficiency of STFSFormer is primarily attributed to its spike-centric computation, including softmax-

free spike attention with spiked $Q/K/V$, associative attention computation that avoids materializing dense attention maps, and block-wise local interactions that constrain attention to short temporal neighborhoods, thereby reducing dense arithmetic and memory traffic on FPGA. Under this setting, STFSFormer achieves 1.2 W average power and 1.5 mJ per EEG segment while maintaining moderate BRAM usage. Compared with the implemented baselines, STFSFormer lowers logic utilization and DSP demand, leading to a clear reduction in per-segment energy without increasing resource budgets, which supports always-on EEG emotion recognition on embedded hardware.

C. Spike Firing Rate Analysis

To assess the biological plausibility and energy efficiency of STFSFormer, we analyze spike firing rates across all HELIF neuron layers on three benchmark datasets. The firing rate is defined as the ratio between the number of spike events and the total number of neuron-time activations, i.e.,

$$\mathrm{FR} = \frac{\sum_\ell \sum_t \sum_i \mathbf{1}[s_i^{(\ell)}(t) = 1]}{\sum_\ell N_\ell T}, \tag{14}$$

where $s_i^{(\ell)}(t) \in \{0, 1\}$ denotes the spike of neuron i in layer ℓ at time step t, N_ℓ is the number of neurons in layer ℓ, and T is the number of time steps.

Across to Table VII, the average firing rates range from $13.85\,\%$ on DREAMER (Arousal) to $17.48\,\%$ on SEED, with DREAMER (Valence) and SEED-V exhibiting $14.62\,\%$ and $16.14\,\%$, respectively. This indicates that the majority of neurons remain inactive at each time step, resulting in sparse, event-driven spiking activity.

Within the multi-frequency spike encoder Φ_{mf}, we further observe a consistent inverse relationship between the decay factor γ and firing rate under identical thresholding and normalization settings. Branches with smaller decay factors ($\gamma_1 = 0.81$), which emphasize high-frequency temporal dynamics, exhibit higher firing rates in the range of $17.26\,\%$–$21.47\,\%$. In contrast, branches with larger decay factors ($\gamma_4 = 0.97$) that capture slower temporal trends produce substantially sparser spike outputs, with firing rates between $8.12\,\%$ and $11.64\,\%$. This behavior is consistent with the intended design of the multi-scale temporal encoding, where fast-varying components elicit more frequent spiking responses, while slow dynamics are encoded more sparsely. Across all tasks, STFSAtten and SpikeMLP layers maintain firing rates below 25%, with deeper layers tending to exhibit progressively sparser representations.

Finally, using a commonly adopted operation-level energy model in which accumulate operations consume approximately $5.1\times$ less energy than MAC operations in 45 nm CMOS technology, we estimate the theoretical energy consumption of STFSFormer relative to equivalent ANNs as

$$\frac{E_{\mathrm{SNN}}}{E_{\mathrm{ANN}}} \approx \frac{\mathrm{FR}}{5.1}. \tag{15}$$

Based on the observed firing rates, this yields an estimated energy ratio of 2.71%–3.43%. While this estimate does not account for memory access or peripheral overheads, it suggests that the sparse spiking activity of STFSFormer can translate into substantial computational energy savings, supporting its suitability for neuromorphic deployment.

TABLE VII: Spike firing rates (%) and theoretical energy consumption across different tasks.

Component	DREAMER		SEED	SEED-V
	Valence	Arousal		
Φ_{mf} (γ_1=0.81)	18.34	17.26	21.47	20.15
Φ_{mf} (γ_4=0.97)	8.75	8.12	11.64	10.42
STFSAtten	15.24	14.56	17.83	16.47
SpikeMLP	16.18	15.42	18.96	17.62
Overall Avg.	14.62	13.85	17.48	16.14
Energy Ratio	2.87%	2.72%	3.43%	3.16%

D. Limitations and Future Directions

Despite the encouraging results, STFSFormer still has limitations. Our evaluation adopts a subject-dependent split and thus does not fully characterize cross-subject/cross-session generalization; future work should test subject-independent protocols and incorporate adaptation to mitigate distribution shifts. In addition, the current block-wise spike-driven attention uses a fixed temporal partition, which could be improved by learnable or content-aware grouping to better handle variable-length and non-stationary affective patterns.

V. Conclusion

In this work, we propose STFSFormer, a spiking Transformer architecture for EEG-based emotion recognition that integrates a high-frequency–enhanced HELIF neuron with a block-wise spike-driven spatiotemporal attention mechanism. Across three datasets, STFSFormer achieves the best accuracy on all evaluated tasks, reaching 92.54 % on SEED, 87.63 % on SEED-V, and 85.91 %/85.47 % on DREAMER valence/arousal. The ablation study further verifies that both the neuron dynamics and the attention design contribute substantially to the final performance. Beyond effectiveness, STFSFormer maintains a favorable efficiency profile, requiring 1.80M parameters and 0.46G MACs with high throughput, and it also demonstrates low energy cost on an FPGA prototype, indicating strong potential for resource-constrained deployment. Overall, STFSFormer demonstrates that co-designing spike dynamics and attention computation around EEG transient time–frequency characteristics enables accurate yet truly always-on affect recognition with measurable hardware-level efficiency gains.

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